DATASCI507: Proposal

Jozef Rivest

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Abstract

Placeholder

1 Introduction

Can political bulletins tell us something about party's level of centralization?

2 Theory

The setup goes as follow. A national party i seeks to gain as much votes as possible in order to win an election. To achieve this, it has to convince the local population across the whole country to i) turnout on the day of the election, and ii) vote for party i. To do this, the party dispose of resources which it can allocates across the different constituencies to help promote certain candidates. The party i also needs to decide how *discretionary* its candidates can be in their campaign. For simplicity, we will treat the allocation of resources as fixed for this model, and focus exclusively on the *centralization* choice of party i.

The problem about centralization goes as follow. The leader of party i does not know about the local needs and issues of all the constituencies. Therefore, by imposing a low discretionary power to the candidates running for its party, he runs the risk to alineate part of the electorate. His priorities might be situated far from those of the local population in constituency d.

3 Data

This project collects an original dataset using political bulletins in Japan for the 2024, 2025, and 2026 lower house elections. The pipeline goes as follow. We first developped a webscrapper to retrieve the election bulletins from go2senkyo.com. The website compiles all the election bulletins across all districts retrieved from each prefecture's electoral commission. The scrapper gets all the PDFs files and store them locally. Once the files are downloaded, another script uses Claude API to segment all the PDFs into the candidates running inside district d. This approach is to facilitate the following step, which is to use PDF parsing and OCR, and to increase the precision of the information retrieved. About the parsing and

the OCR, we used the Yomitoku Python librairy that is optimized for Japanese text and documents.

Once the parsing is done, we can store the information into a Pandas dataframe. Each candidates (rows) have the following information attached to them: 1) text from their election bulletin, 2) the number of paragraphs, 3) the number of headings, 4) the total number of characters (including both Kanji and Kana).